

NEPHROTIC SYNDROME

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Outline

- Definitions
- Classification
- Epidemiology
- Pathogenesis
- Clinical features
- Treatment
- Prognosis

Nephrotic Syndrome (NS)

- Definitions
- NS
 - A *syndrome* characterized by:
 - *Proteinuria*
 - *Hypoalbuminemia* ($<3\text{g/dl}$)
 - Edema and
 - Hyperlipidemia(cholesterol $> 200\text{ mg/dL}$).
 - Nephrotic range proteinuria – urinary excretion $> 1\text{ g/m}^2/\text{day}$ (40 $\text{mg/m}^2/\text{hr}$) or a first morning protein : creatinine ratio of $>2:1$

Nephrotic syndrome

- Primarily a pediatric disorder
- 15 times more common in children than adults.



Classification

❖ Etiologic

➤ Primary

- associated with primary glomerular disease

➤ Secondary

- Occurs as part of recognized systemic disease or results from some evident cause; possible causes are as follows:
- Post infectious (G-A β -hemolytic strep., Syphilis, malaria, TB and viruses)
- Collagen vascular disease (SLE, RA, polyarteritis nodosa)
- Drugs (mercury, gold, penicillin, captopril...)

Classification...

- HSP, sickle cell disease, DM, malignancy (leukemia, lymphoma, wilm's tumor)

❖ **Histologic**

- Minimal lesion NS
- Diffuse mesangial hypercellularity
- Focal glomerulosclerosis
- Membranous glomerulonephritis
- Membranoproliferative glomerulonephritis

Table 545.1

Causes of Childhood Nephrotic Syndrome

IDIOPATHIC NEPHROTIC SYNDROME
Minimal change disease
Focal segmental glomerulosclerosis
Membranous nephropathy
Glomerulonephritis associated with nephrotic syndrome—membranoproliferative glomerulonephritis, crescentic glomerulonephritis, immunoglobulin A nephropathy
GENETIC DISORDERS ASSOCIATED WITH PROTEINURIA OR NEPHROTIC SYNDROME (see also Table 545.3)
<i>Nephrotic Syndrome (Typical)</i>
Finnish-type congenital nephrotic syndrome (absence of nephrin)
Focal segmental glomerulosclerosis (mutations in nephrin, podocin, <i>MYO1E</i> , α -actinin 4, TRPC6)
Diffuse mesangial sclerosis (mutations in laminin β_2 chain)
Denys-Drash syndrome (mutations in WT1 transcription factor)
Congenital nephrotic syndrome with lung and skin involvement (integrin α -3 mutation)
Mitochondrial disorders

SECONDARY CAUSES OF NEPHROTIC SYNDROME

Infections

Endocarditis

Hepatitides B, C

HIV-1

Infectious mononucleosis

Cytomegalovirus

Malaria

Syphilis (congenital and secondary)

Toxoplasmosis

Tuberculosis

Schistosomiasis

Filariasis

Drugs

Epidemiology

- Incidence 1-3 per 100,000 children/year , < 16 yr of age.
- Primary NS accounts for approximately 90% of all cases
- Age of onset and sex incidence varies with the type of NS
- The most common type of NS in children is **Minimal lesion**
NS

Glomerular handling of proteins

- ❖ Glomerular capillary wall has three components:
 - The fenestrated endothelial cell
 - The glomerular basement membrane (GBM)
 - The epithelial cell foot processes - the pores between the epithelial foot processes are closed by a thin membrane called the slit diaphragm

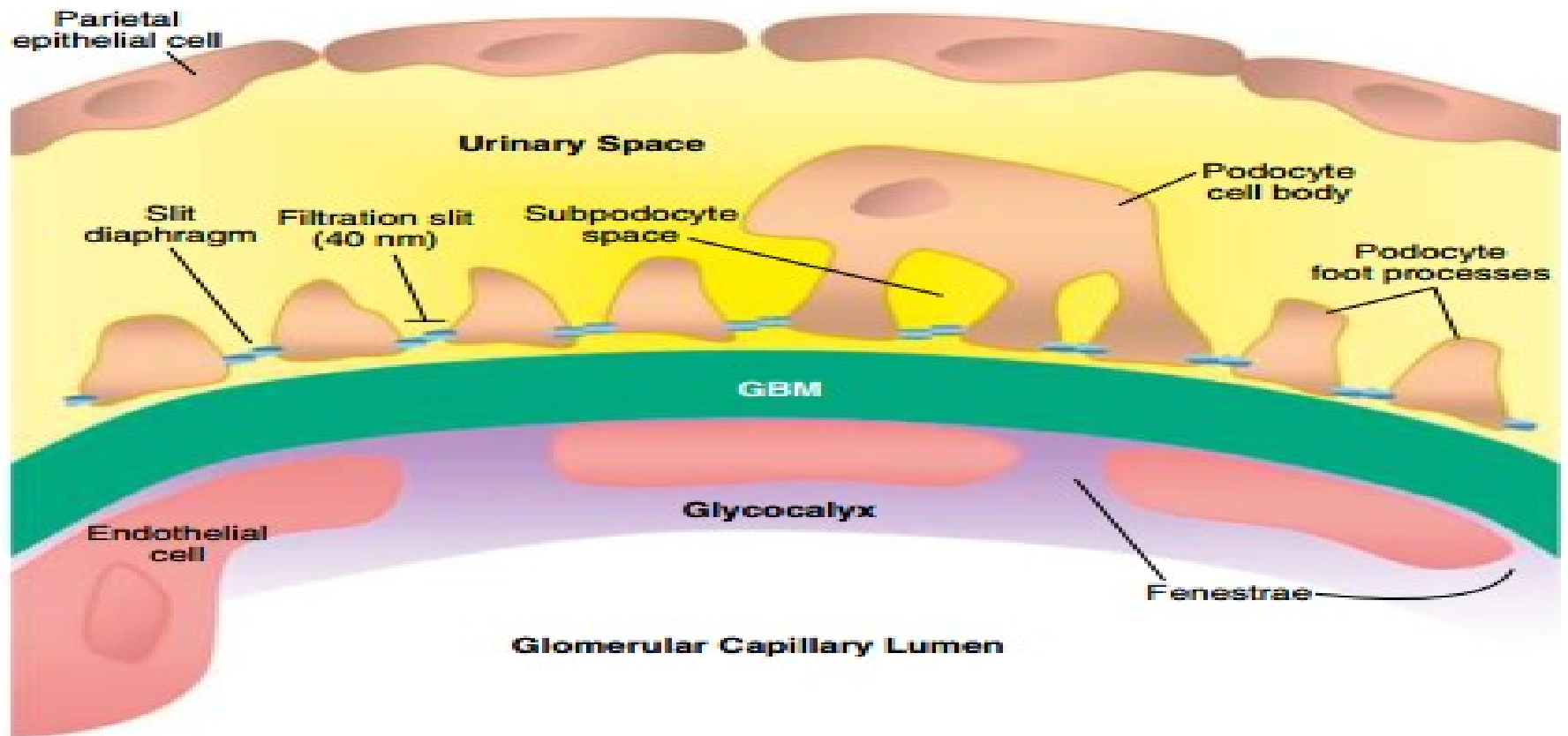
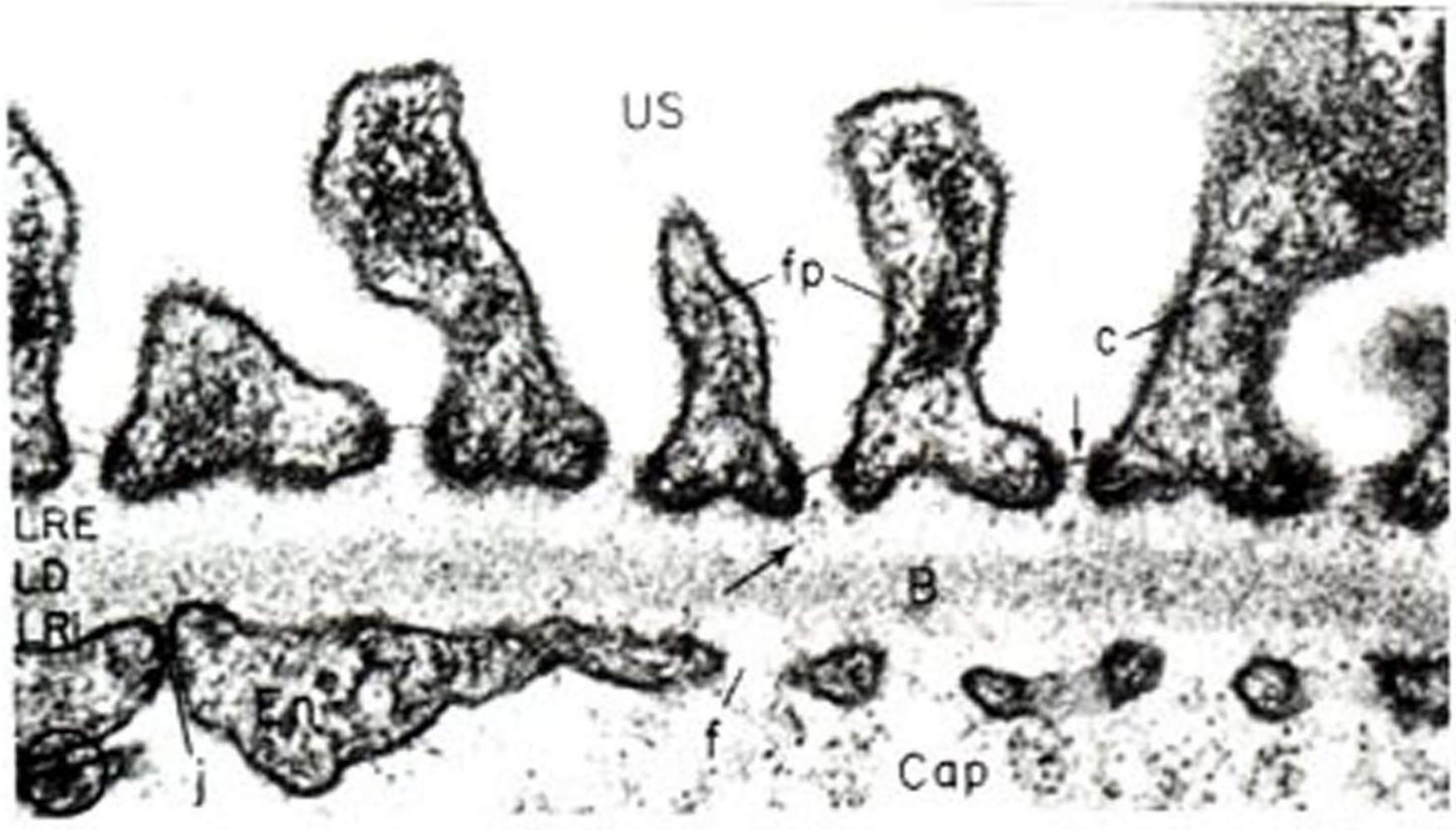


Fig. 566.2 Glomerular capillary wall. The three layers of the capillary wall (glomerular endothelial cell, glomerular basement membrane [GBM], and podocyte) act as the glomerular filtration barrier (GFB), preventing proteins and large molecules from passing from the capillary lumen into the urinary space. The podocyte cell body lies within the urinary space, and the cell is attached to the GBM through foot processes. Adjacent foot processes are separated by the filtration slit, bridged by the slit diaphragm. Disruption of the GFB leads the passage of protein across the capillary wall, leading to proteinuria. (From Jefferson JA, Nelson PJ, Najafian B, et al. Podocyte disorders: core curriculum 2011. *Am J Kidney Dis.* 2011;58:666–677. Fig. 1.)

Glomerular capillary wall



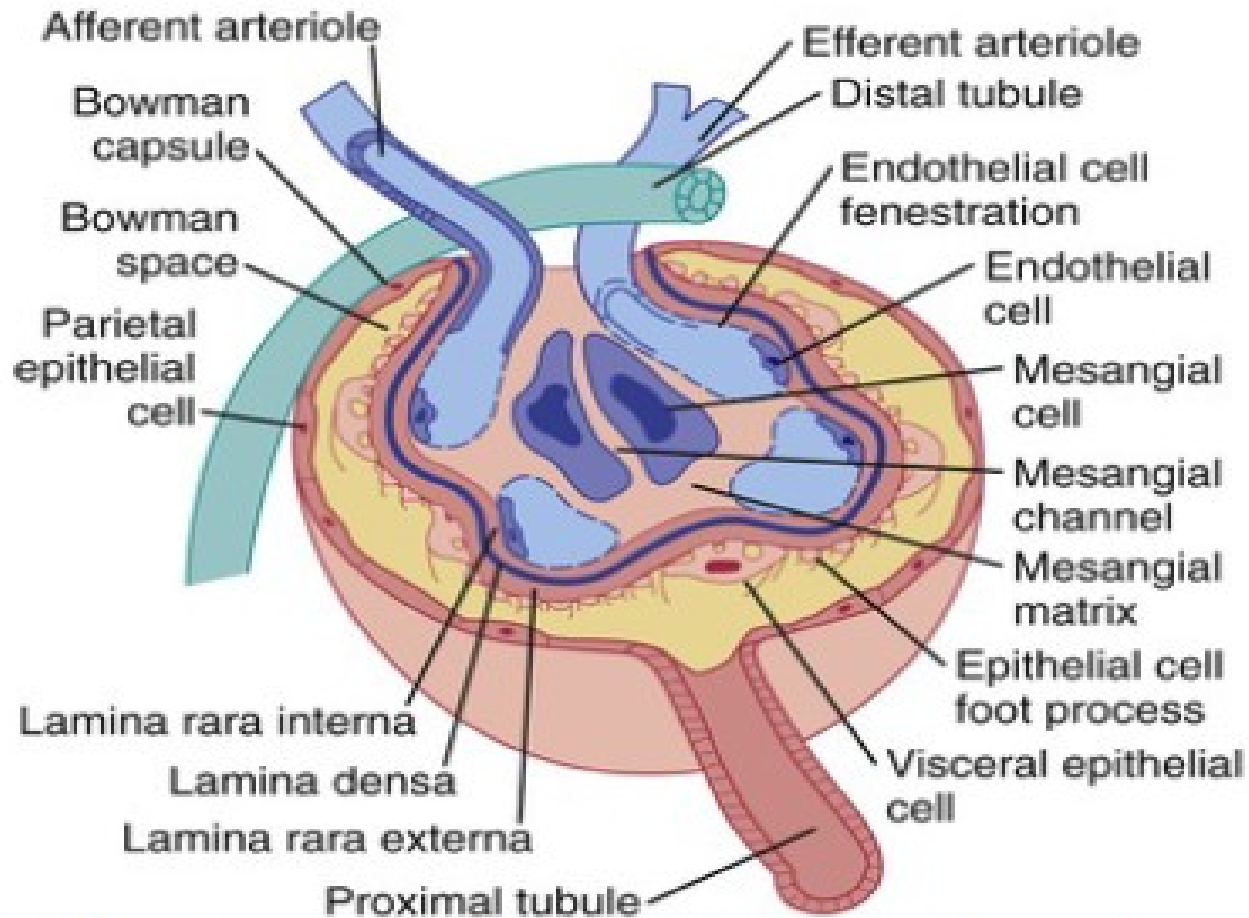


FIG. 535.4 Schematic depiction of the glomerulus and surrounding structures.

Glomerular handling.../2

Filtration barrier by:

- **Size** (albumin “not allowed”) : the endothelial cell is not part of the size barrier
- **Charge** (*heparan sulphate glycosaminoglycan*), the endothelial cell and GBM are negatively charged
- Slit diaphragm proteins (nephrin, podocin, etc.)
Essential for prevention of protein filtration

Mechanisms of **increased protein** excretion:

Glomerular proteinuria

- Commonest cause in children
- Due to increased filtration of macromolecules like albumin
- May be due to anatomic or functional lesions
- The cause might be fever, exercise, or orthostatic

Mechanisms of **increased protein** excretion:

Tubular proteinuria:

- Due to reduced reabsorption of low molecular weight proteins in the proximal tubule
- Beta-2 and alpha-1 microglobulins and retinol binding protein are excreted
- usually associated with the Fanconi syndrome, glycosuria, phosphaturia, PRTA, cystinuria...or tubulointerstitial nephritis

Mechanisms of **increased protein** excretion:

Overflow proteinuria

- Due to increased production of a particular LMWP to a level beyond the absorptive capacity of the proximal tubule.
- Increased production of Ig light chains in multiple myeloma, usually in adults is an example

Idiopathic NS

- Accounts for approximately 90% of NS cases in b/n 1-10 years & 50% after 10 years
- It occurs in three morphologic patterns:
 - Minimal change disease (85%)
 - Focal segmental glomerulosclerosis (10%)
 - Mesangial proliferative (5%)
- Occurs between 2-6 yrs commonly
 - But is reported as early as 6 months and throughout adulthood
- Male to female ratio is $\sim 2:1$

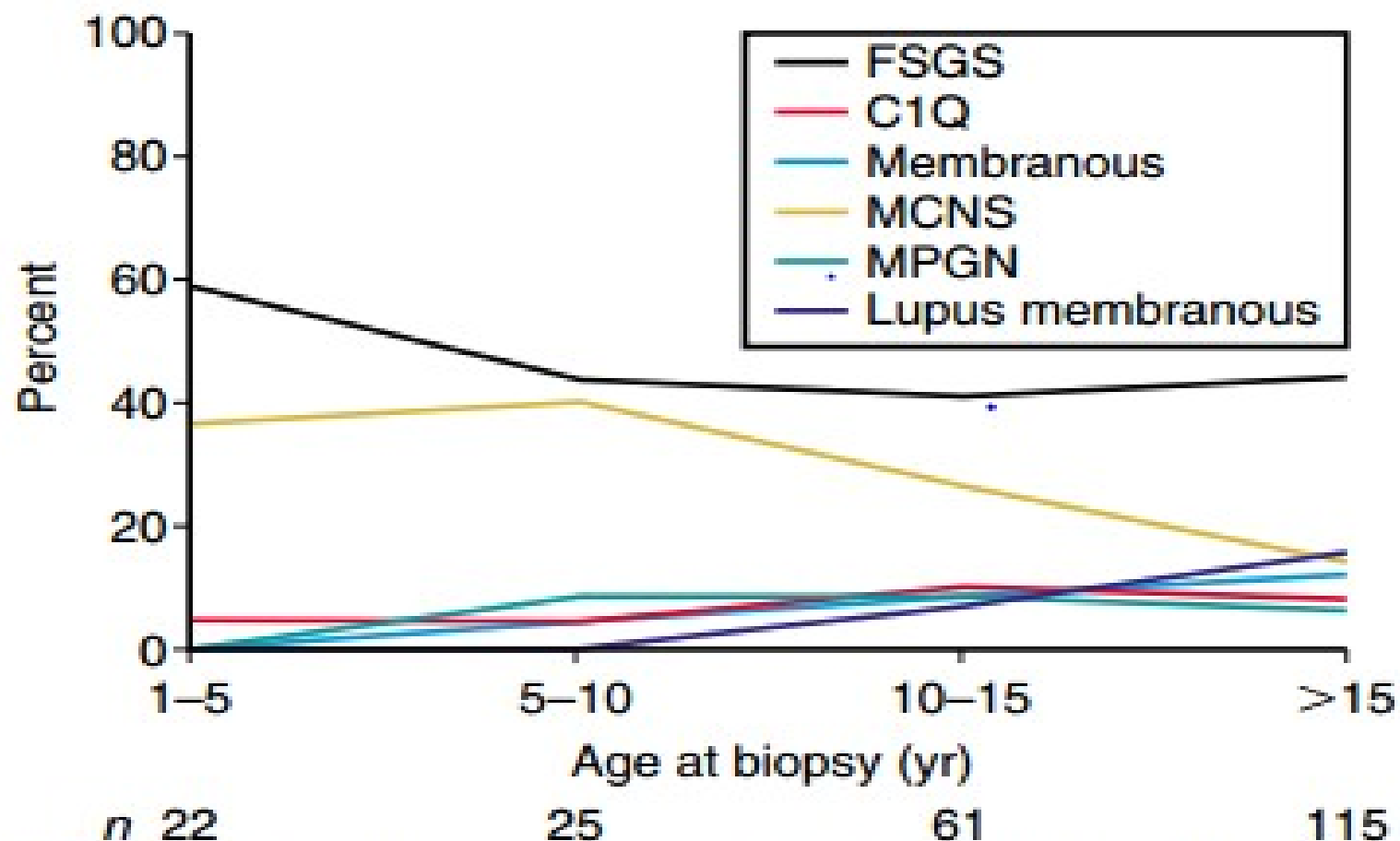


Fig. 567.1 Kidney biopsy results from 223 children with proteinuria referred for diagnostic kidney biopsy (Glomerular Disease Collaborative Network, J. Charles Jennette, MD, Hyunsook Chin, MS, and D.S. Gipson, 2007). C1Q, Nephropathy; FSGS, focal segmental glomerulosclerosis; MCNS, minimal change nephrotic syndrome; MPGN, membranoproliferative glomerulonephritis; n, number of patients. (From Gipson DS, Massengill SF, Yao L, et al. Management of childhood onset nephrotic syndrome. *Pediatrics*. 2009;124:747-757.)

Idiopathic...

- **Pathphysiology**

- Results from alteration in the permeability selective characteristics of the GBM
- Protein loss is generally $> 2\text{g}/24\text{ hr}$ (normal $<150\text{ mg}/24\text{ hr}$)
- Protein lost is primarily albumin resulting in hypoalbuminemia
- Edema appears when serum albumin falls below 2.5 g/dl
- Normal protein excretion in children $< 100\text{ mg}/\text{m}^2/\text{day}$ or $<4\text{ mg}/\text{m}^2/\text{hr}$. or a total of $150\text{ mg}/\text{d}$, in the neonates up to $300\text{ mg}/\text{d}$

- Pathophysiology.....

The **normal low rate** of protein excretion is due to:

- Restriction of filtration of proteins across the GBM
- Reabsorption of the filtered proteins by the proximal tubule

Table 545.2**Summary of Primary Renal Diseases That Manifest as Idiopathic Nephrotic Syndrome**

FEATURES	MINIMAL CHANGE NEPHROTIC SYNDROME	FOCAL SEGMENTAL GLOMERULOSCLEROSIS	MEMBRANOUS NEPHROPATHY	MEMBRANOPF GLOMERULON Type I
DEMOGRAPHICS				
Age (yr)	2-6, some adults	2-10, some adults	40-50	5-15

Sex (male:female)	2 : 1	1.3 : 1	2 : 1	1:1
CLINICAL MANIFESTATIONS				
Nephrotic syndrome	100%	90%	80%	60%*
Asymptomatic proteinuria	0	10%	20%	40%
Hematuria (microscopic or gross)	10-20%	60-80%	60%	80%
Hypertension	10%	20% early	Infrequent	35%
Rate of progression to renal failure	Does not progress	10 yr	50% in 10-20 yr	10-20 yr
Associated conditions	Usually none	HIV, heroin use, sickle cell disease, reflux nephropathy	Renal vein thrombosis; medications; SLE; hepatitis B, C; lymphoma; tumors	None

GENETICS				
	None except in congenital nephrotic syndrome (see Table 545.3)	Podocin, α -actinin 4, TRPC6 channel, INF-2, MYH-9	None	None
LABORATORY FINDINGS				
	Manifestations of nephrotic syndrome ↑ BUN in 15–30% Normal complement levels	Manifestations of nephrotic syndrome ↑ BUN in 20–40% Normal complement levels	Manifestations of nephrotic syndrome Normal complement levels	Low complement levels—C1, C4, C3-C9
RENAL PATHOLOGY				
Light microscopy	Normal	Focal sclerotic lesions	Thickened GBM, spikes	Thickened GBM, proliferation
Immunofluorescence	Negative	IgM, C3 in lesions	Fine granular IgG, C3	Granular IgG, C3
Electron microscopy	Foot process fusion	Foot process fusion	Subepithelial deposits	Mesangial and subendothelial deposits
REMISSION ACHIEVED AFTER 8 WK OF ORAL CORTICOSTEROID THERAPY				
	90%	15–20%	Resistant	Not established/resista

* Approximate frequency as a cause of idiopathic nephrotic syndrome. Approximately 10% of cases of adult nephrotic syndrome are a result of various diseases that usually manifest as acute glomerulonephritis.

Approach to the infant or child with nephrotic syndrome

Initial history and physical examination focused on potential causes or complications of NS*

Initial laboratory evaluation: ¶

- First morning void to measure urinary protein to creatinine ratio or 24-hour urine collection
- Serum albumin
- Urinalysis with microscopy
- Serum electrolytes
- BUN and creatinine
- Lipid panel
- Complement levels

Are there any of the following:

- Age <3 months
- Extra-renal features suggestive of a congenital syndromeΔ
- Family history suggestive of hereditary nephrotic syndrome

Yes

- Likely congenital or genetic NS
- Refer for genetic testing

No

Are any of the following present:

- Hypertension
- Gross hematuria
- Elevated serum creatinine
- Low serum complement levels
- Extra-renal symptoms suggestive of a secondary cause◇

Yes

- Causes other than MCD are likely§
- Perform kidney biopsy to identify specific cause
- Additional evaluation may include testing for specific diseases, depending on presentation

No

What is the patient's age?

3 months to 12 years

- MCD most likely cause
 - Trial course of glucocorticoids‡
- Does the patient respond?

Yes

MCD diagnosis supported

No

- Refer to other content on steroid-resistant nephrotic syndrome
- Additional workup may include genetic testing and kidney biopsy, if not already performed

>12 years

- Causes other than MCD are likely§
- Perform kidney biopsy to identify specific cause

Idiopathic...

- Serum level of almost all lipids is increased as a result of increases in lipoprotein synthesis and low levels of lipoprotein lipase
- Clinical Manifestations
 - Attacks may follow URTI
 - Usually presents
 - With edema, initially noted in periorbital area and lower extremities
 - Edema becomes generalized with time
 - Some may present with hypotension

Atypical features

- Infants and children >10yrs
- Positive family history of NS
- Presence of extra renal symptoms like rash, arthritis, anemia
- Hypertension
- Renal insufficiency
- Gross hematuria

Idiopathic...

- Rarely may develop renal failure as a result of the hypotension
- With massive pleural effusion and ascites, patients may develop dyspnea
- Unlike nephritic syndrome, gross hematuria and hypertension are uncommon

➤ **Lab findings**

- Urinalysis reveals proteinuria of +3 or +4 on dipstick
- Serum albumin - <3 g/dl
- High serum cholesterol and triglyceride level

Idiopathic...

- Normal C3 level , electrolytes, RFT, CBC
- Renal biopsy - is indicated in children with:
 - Macroscopic hematuria, hypertension, renal insufficiency, hypocomplementemia, whose age is **<1 yr** or **>12yr**, extra-renal symptoms (ie, malar rash, purpura, adenopathy), steroid resistant

Complications

- Infection is the major complication
 - Commonest infection is spontaneous peritonitis
 - Commonest cause – strep. Pneumoniae
 - Other infections include
 - Pneumonia, cellulitis, sepsis and UTI
- Thromboembolism
- Renal insufficiency
- Anasarca
- Hypovolemia....

DDx for edematous child

- Protein losing enteropathy
- Hepatic failure
- Heart failure
- SAM
- Acute or chronic GN

Idiopathic...

➤ Treatment

- *The goal of therapy should be a gradual reduction of edema with judicious use of diuretics, sodium restriction, and cautious use of intravenous albumin infusions, if indicated.*
- 1. General measures: focused on preventing or minimizing complications
 - Children with mild to moderate edema can be managed as outpatient
 - Salt restriction
 - Diuretics can be used cautiously to reduce severe form of edema:
 - Frusemide – respiratory distress due to massive pleural effusion and ascites or severe scrotal edema
 - Careful monitoring of RF and electrolytes

Treatment...

- **Infection control**

- ✓ Prevention of infection with vaccination with 23-valent polysaccharide ([PPSV23](#)) pneumococcal and varicella vaccines if not previously immunized,
- ✓ live vaccinations should be avoided during relapse and while on daily immunosuppressive medications.
- ✓ Identification and treatment of suspected infections

Treatment...

2. Steroids

- Consideration:..
 - Prednisolone – given in a dose of $60 \text{ mg/m}^2 / 24 \text{ hr}$ or $2 \text{ mg/kg}/24 \text{ hr}$ (Max. $60 \text{ mg}/24 \text{ hr}$) in 2 divided dose for at least 4 wks
 - 80-90% of patients respond in a median time of 10 days to remission
 - After the initial 4-6 wk course, Prednisolone should be tapered to $40 \text{ mg/m}^2/24 \text{ hr}$ every other day every morning

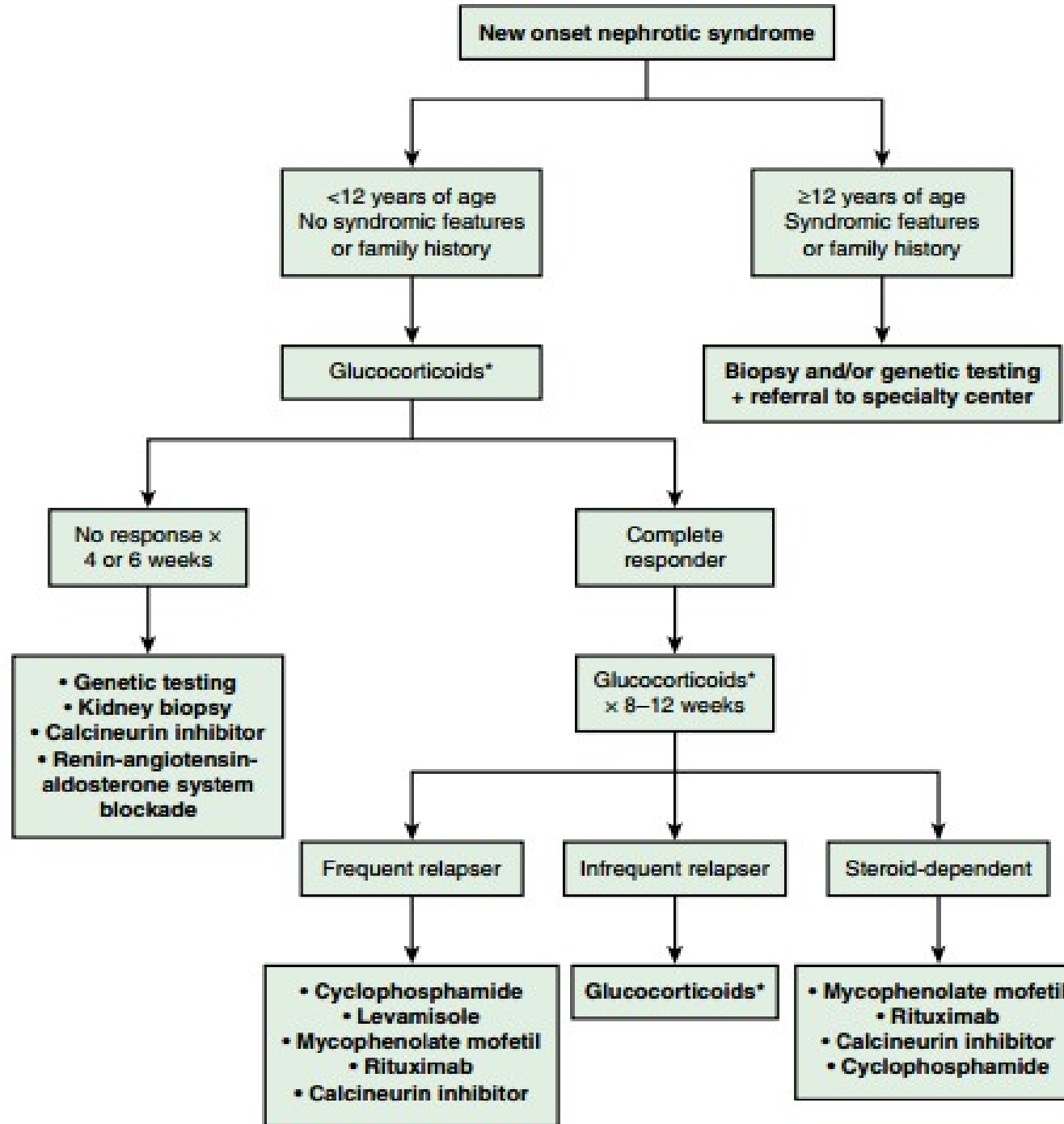


Fig. 567.4 Treatment algorithm for new-onset nephrotic syndrome (NS). Therapeutic approach to NS in children from onset. *Glucocorticoids include PO prednisone or prednisolone. NS, Nephrotic syndrome. (From *Kidney Disease: Improving Global Outcomes (KDIGO) Glomerular Diseases Work Group. KDIGO 2021 Clinical Practice Guideline for the Management of Glomerular Diseases. Kidney Int. 2021;100[4S]:S1–S276. Fig. 40.*)

Treatment...

- The alternate day dose is slowly tapered over the next 2-3 months
- Response to steroids
 - >95% with minimal change NS
 - ~50% with mesangial proliferation NS
 - ~20% with focal glomerulosclerosis NS
- Cytotoxic drugs
 - Cyclophosphamide – used in frequently relapsing NS or in severe steroid toxicity or in steroid resistant NS

Treatment response

- **SRNS** is NS resistant to steroid therapy, defined by the absence of complete remission after an initial course of daily [prednisone](#) therapy at a dose of 60 mg/m² per day.
- **Complete remission** is resolution of proteinuria defined as a UPCR based on a first morning void or 24-hour collection of ≤ 0.2 mg prt/mg cr or urine dipstick readings of negative or trace protein on three consecutive days.
- **Partial remission** is a decrease in proteinuria with a UPCR based on first morning void or 24-hour urine sample of >20 but <200 mg/mmol and, if available, serum albumin ≥ 30 g/L.
- **Relapse** is the recurrence of nephrotic-range proteinuria following either

Definitions...

- Steroid responsive NS
 - Disappearance of the proteinuria within 8 wks of starting treatment with a corticosteroid
- Steroid resistance NS
 - When there is no remission after 8 wks of treatment with steroid
- Steroid dependent NS
 - relapsing during taper or within two weeks of discontinuation of steroid therapy

Definitions...

- Remission of NS
 - When urine is negative or trace for protein for a period of at least 3 days
- Relapse of NS
 - The occurrence of proteinuria measuring 2+ or more on 3 consecutive days in a patient who is in remission
- Frequently relapsing course
 - Greater than 2 relapse of NS with in 6 months

Idiopathic...

- Prognosis
 - Most children with steroid responsive NS have repeated relapses till the end of 2nd decade

References

- **Nelsons 22 edition**
- **Up To Date**

- **THANK YOU**